

Self-Test

- In this self test you will write a Contact management program that can allow you to manage user contact information. This program will demonstrate your knowledge of Generic classes and ArrayLists.
- Create a class named **Contact**. This class **implements Comparable** and has three (3) instance variables, **String name**, **String telNum** and **String email**.
- Create two constructors.
 - The first constructors accepts parameters for all the instance variables.
 - The second constructor accepts only the **name**. the other varuables are set to null.
- Create accessor and mutator methods for the following **getTelNum()**, **setTelNum()**, **getEmail()**, **setEmail()**, **getName()**. {use the appropriate parameters}
- Create a toString() method that returns a String containing the name, telNum and the email.
- Create an equals() method that determines equality by using the name.
- Create a compareTo() method that compares using the name

Self-Test

- Create another class named **GenericManager**. This class has a single generic parameter T. The generic parameter T **extends** Comparable.
- This class has a single instance variable ArrayList **cList**. The base type of this ArrayList is T.
- Create a constructor that accepts no parameters. The constructor creates a new ArrayList object.
- Create a method **add()**. **This method accepts a generic type as a parameter** and adds it to the ArrayList. It then returns void.
- Create a method **sort()**. This method sorts the ArrayList and returns void.
- Create a method **find()**. **This method accepts a generic type as a parameter** and returns a String containing all the objects in the list that are equal to the parameter.
- Create a method **toString()**. This method goes through the ArrayList item-by-item and returns a String of all the objects in the ArrayList.

Self-Test

- Test your classes with the following code

```
import java.util.InputMismatchException;
import java.util.Scanner;

public class test {

    public static void main(String[] args){

        GenericManager<Contact> contactManager = new GenericManager<Contact>();

        boolean exit = false;
        Scanner keyboard = new Scanner(System.in);
        do{
            //print the menu
            System.out.println("*****Contact Management System*****");
            System.out.println("(1) Add a contact: ");
            System.out.println("(2) Display all contacts: ");
            System.out.println("(3) Search for contacts");
            System.out.println("(4) Sort the contacts");
            System.out.println("(5) Exit the program");
            System.out.println("*****");

            //using an Exception control loop
            try{
                int input = keyboard.nextInt();

                switch(input){


```

```
                    case 1:
                        System.out.println("Please enter the name:");
                        String name = keyboard.next(); name+=keyboard.nextLine();
                        System.out.println("Please enter the tell#");
                        String tel = keyboard.next();
                        System.out.println("Please enter the email:");
                        String email = keyboard.next();
                        contactManager.add(new Contact(name,tel,email));
                        break;
                    case 2:
                        System.out.println("*****Contact List*****");
                        System.out.println(contactManager.toString());
                        System.out.println("*****");
                        break;
                    case 3:
                        System.out.println("Please enter the name to search for:");
                        String aname = keyboard.next(); aname+=keyboard.nextLine();
                        System.out.println("*****Search Results*****");
                        System.out.println(contactManager.find(new
                            Contact(aname)));
                        System.out.println("*****");
                        break;
                    case 4:
                        contactManager.sort();
                        break;
                    case 5:
                        exit = true;
                        System.out.println("Exiting Program... Goodbye.");
                        break;
                    default: break;
                }
            }catch(InputMismatchException e){
                keyboard.nextLine();
                System.out.println("Incorrect input, Try Again.. \n");
            }
        }while (!exit);

    }
}
```

Self-Test

- Sample Output

```
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```

```
1
Please enter the name:
John Doe
Please enter the tell#:
01022334567
Please enter the email:
jdoo@gmail.com
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```

```
1
Please enter the name:
Jane Doe
Please enter the tell#:
01022334567
Please enter the email:
janeD@gmail.com
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```

```
2
*****Contact List*****
Name: John Doe
Tel #: 01022334567
Email: jdoo@gmail.com

Name: Jane Doe
Tel #: 01022334567
Email: janeD@gmail.com

*****
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```

```
2
*****Contact List*****
Name: Jane Doe
Tel #: 01022334567
Email: janeD@gmail.com

Name: John Doe
Tel #: 01022334567
Email: jdoo@gmail.com

*****
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```

```
3
Please enter the name to search for:
John Doe
*****Search Results*****
Name: John Doe
Tel #: 01022334567
Email: jdoo@gmail.com

*****
*****Contact Management System*****
(1) Add a contact:
(2) Display all contacts:
(3) Search for contacts
(4) Sort the contacts
(5) Exit the program
*****
```